ECSE 552 Final Project Proposal: CARLA - Catan Advanced Reinforcement Learning Agent

Background

Settlers of Catan is a four-player board game in which players gather resources from dice rolls, build on a randomly generated hexagonal board, and trade with the bank and each other to reach 10 victory points first. It is stochastic, partially observable (hidden hands and development cards), mixes competition with cooperation (trading), and has a long horizon and a huge branching factor.

Application. The proposed project is to build a deep reinforcement learning (RL) agent that plays the full game, including player-to-player trading. The agentic system will be called Catan Advanced Reinforcement Learning Agent (CARLA). Comparable games have been mastered with search plus learned evaluation [1] and large-scale self-play PPO [2]. Catan's mix of hidden information, chance and negotiation mirrors real-world problems such as multi-party negotiation, resource allocation under uncertainty, and automated trading.

Problem formulation. Catan will be modelled as a partially observable stochastic game: the agent sees its own hand and public information, selects one legal action per decision, and must maximise its win probability against fixed opponents. Two gaps motivate a solution. First, prior work has implemented learning for four-player Catan with player-to-player trading, but has not demonstrated reliable learned trading alongside strong performance across the full game [3]. Other approaches drops players, drops trading, or hand-code most decisions [4, 5]. CARLA will test whether feasibility-aware trade actions and opponent-aware learning improve trading and performance. Second, terminal win/loss feedback is sparse over a long game. Dense rewards for individual actions can help assign credit, but may favor proxies such as gaining points or making trades without improving win probability. The project will test whether potential-based shaping using the agent's score margin over its strongest opponent improves learning speed or final win rate, compared with sparse win/loss and own-points shaping.

Aims/Goals

Design constraints

1. **Legality:** every sampled action is legal, which requires per-head masks conditioned on earlier heads.
2. **Fair information:** the policy sees only its seat's information; opponents' true hands may appear only in training losses.
3. **Symmetry:** the policy should be equivariant to board rotations/reflections and player-seat permutations: transforming the state should transform the action distribution correspondingly. Value estimates should be invariant. A coordinate-free graph encoder and relative seats will support this structure.
4. **Reward alignment:** shaping must be potential-based to preserve optimal policies [6], must not reward proxies (trade spam, hoarding), and must measure advantage *relative* to opponents.
5. **Robustness to chance:** dice and card draws dominate single games, so results need many games, rotated seats and fixed seeds.
6. **Performance:** $\geq$ 99% win rate against random bots and a statistically significant margin over the heuristic bot in four-player tournaments. An extreme rate increase against a competent heuristic bot is unlikely, since dice luck lets a decent amateur player win a sizeable share of games against any opponent.

Scope. The proposed project will test whether opponent-aware reward shaping improves learning and final strength, focusing on trading and long-horizon credit assignment. The comparison of (i) Sparse win/loss, (ii) own-points shaping, and (iii) opponent-aware shaping (whose potential is the agent's point margin over its strongest opponent) will drive the study. A secondary aim is the encoder and belief ablation below. The project's scope is the full base game with maritime trading and switchable domestic trading. Out of scope: Catan expansions, natural-language negotiation and human evaluation.

Methodology

Proposed solution. CARLA plays the full four-player game, including maritime and domestic trading, and is trained with PPO [7] over a factored action space. An action-type head comes first, then type-specific heads (intersection, edge, tile, victim, development card, resources) and a trade decoder. Each head is masked given earlier heads, so illegal actions are impossible. Unlike SettlersRL [3], the board (19 tiles, 54 intersections, 72 edges) is a coordinate-free graph processed by a GNN [8], making the policy invariant to board symmetries. Opponents, development cards and trade offers are encoded with self-attention [9]. A GRU belief module summarises public events and gets an auxiliary loss on opponents' true hands, which the simulator provides during training.

Implementation plan. Our implementation plan features (1) Engine: a from-scratch, seeded Python simulator of the full rules with legal-action generation and exact replay, already complete and tested. (2) Learning: CPU workers run self-play against a pool of past snapshots, random and heuristic bots; a GPU learner performs PPO with GAE, after a behavioural-cloning warm start on heuristic games. The three reward formulations are interchangeable config options, trained with otherwise identical settings. (3) Architecture: MLP baseline (implemented), GNN, and GNN with attention encoders, plus a rule-based or learned belief, selected by configuration for controlled ablations.

Validation, testing, and evaluation. Engine tests check rule invariants and replay hashes over randomized games; a leakage test confirms observations ignore opponents' hidden state. Agents play tournaments against fixed bots; we report win rates with 95% confidence intervals over at least three seeds [10] (about ± 3 points at 1,000 games). The main experiments are the reward comparison and an ablation over encoder, belief and domestic trading. We also track trade acceptance and rotation invariance.

Alternative solutions. Alternative solutions to playing Catan have been explored. Heuristic or search-based players [11, 12] depend on hand-engineered knowledge. Earlier learning approaches simplify the problem: [4] restrict play to two players without trading, and [5] learn only trading while a heuristic makes every other decision. The closest prior work, SettlersRL [3], has several limitations: it uses a flat tile encoder, does not mask infeasible trades, uses no demonstrations and is evaluated mostly against its own checkpoints. CARLA addresses each: masked, feasibility-aware trades; behavioural-cloning warm starts; a symmetry-respecting graph encoder, which should improve sample efficiency on our compute budget; and evaluation against fixed baselines with confidence intervals.

Feasibility and resources

No external dataset is needed: the simulator generates unlimited games, heuristic games provide demonstrations, and held-out seeds form the test set. The codebase already has the engine (about 200 tests), random and heuristic bots, environments, rewards, rollout collection, the MLP baseline, tournaments with confidence intervals and a GUI; domestic trading is under review. Still to build: PPO updates, the opponent pool, opponent-aware rewards, the GNN, attention and belief modules, and behavioural cloning. A Ryzen 9800X3D workstation with an RTX 5080 runs about 560 games/s on 8 cores, and a GPU Slurm cluster is available for seed sweeps.

Risk analysis

The main risk is that learning stalls under the sparse reward or that self-play cycles between strategies; we mitigate this with the behavioural-cloning warm start, reward shaping, and fixed bots kept in the opponent pool. Shaping itself may be exploited, for example by chasing points instead of winning, so it is potential-based and always compared against a sparse-reward baseline. Trading may degenerate into spam or disappear entirely, which feasibility masks, per-turn offer limits and trade metrics will reveal. If compute proves insufficient, we prioritise the MLP reward study and shrink the ablation grid after pilot runs. Finally, engine bugs or information leakage could silently corrupt results; invariant, replay and leakage tests run in CI, and games can be inspected visually in the GUI.

Role of group members

All four members contribute to every part of the codebase rather than owning fixed sections. Work is organised on GitHub: each task (a feature, experiment or fix) is developed on its own branch and merged through a pull request that another member reviews and that must pass continuous integration (tests, linting and type checks). We meet online at least once a week to track progress and discuss important decisions. The final report will be written the same way, with sections drafted and reviewed collaboratively instead of being assigned to individuals.

References

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